



Delivering the ‘Whole Product’: Business Model Impacts and Agility Challenges in a Network of Open Source Firms

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ABSTRACT

It has been argued that competitive necessities will increasingly require OSS companies to participate in cooperative business networks in order to offer the complete product/service (whole product) demanded by customers. It is envisaged that these business networks will enhance the business models of participant firms by supplementing their value-adding activities and increasing responsiveness to customers. However, while such propositions have intuitive appeal, there is a lack of empirical research on such networks. This article examines Zea Partners, a network of small open source companies cooperating to deliver the ‘whole product’ in the area of content management systems (CMSs). It investigates how participation in the network augments the business models of participant companies and identifies the business agility challenges faced by the network. The article concludes that reconciling the coordination needs of OSS networks with the operational practices of participant firms is a critical issue if such networks are to achieve adaptive efficiency to deliver whole products in a ‘bazaar-friendly’ manner.

Keywords: agility; business models; business networks; open source software

INTRODUCTION

Researchers (Agerfalk, Finnegan, Hayes, Lundell, & Ostling, 2006; Feller, Finnegan, Hayes, & Lundell, 2006a; Fitzgerald, 2006) have recently argued that open source software firms should adopt a ‘whole product’ approach (cf. Moore, 1999) by forming a network/ecosystem of partners with complementary capabilities “to offer a professional product and service in an agile, bazaar-friendly manner” (Fitzgerald,

2006, p. 294). This ‘whole product’ approach is consistent with the challenges of “productizing OSS” discussed by Woods and Guliani (2005), as well as developments in the production and use of other complex product/service offerings as discussed by Davidow and Malone (1992). This approach is regarded as appropriate when there is a need for firms to quickly deliver a variety of customized products, and when the nature of the product development process

means that individual organizations do not have sufficient competencies to deal with all parts of product design (Davidow & Malone, 1992; Huang, 2001). In such circumstances, market forces require organizations with similar goals to align themselves in IT-mediated partner networks in order to meet customer requirements (Stafford, 2002).

Moore (1999) popularized the concept of the 'whole product' as the cornerstone of market-driven, rather than product-driven, businesses. However, the concept resonates with the dynamics of the open source software phenomenon, which, due to the licensing structure, emphasizes services and meta-services surrounding the artifact. Indeed, Woods and Guliani (2005) describe as the challenge of 'productizing' open source software as the need to offer support, implementation, modification, and related services. Thus, networks of cooperating, small, open source software organizations may represent what Clemons and Row (1992) term a "move-to-the middle," where networks of organizations interact in order to deliver value (in the form of the whole product) to the end consumer.

This article examines Zea Partners, a business network of firms developing content management systems and selling related services, all based around the Zope application server. It investigates how participation in the network augments the business models of participant firms in order to adopt a 'whole product' approach, and identifies the challenges faced by the network in trying to ensure the business agility necessary to offer the 'whole product'. The article begins by discussing the theoretical foundation for the study. Next, the research objective and research methods are discussed. The case environment is then outlined and the findings presented. The article illustrates that participation in the network allows firms to share business model components within a centrally managed network, and to engage in agile competitive practices by making network-level changes in response to changes in the external environment. The need to address adaptability and alignment issues in addition to

business agility is highlighted, however. Consequently, the article concludes that reconciling the coordination needs of OSS networks with the operational practices of participant firms is a critical issue if such networks are to achieve adaptive efficiency to deliver whole products in a bazaar-friendly manner.

THEORETICAL FOUNDATION

OSS has been investigated from a variety of disciplinary and theoretical perspectives. The two dominant research themes, however, have been: (1) OSS software engineering tools and techniques and (2) the socio-cultural analysis of OSS communities. Commercial organizations are under-represented in the research, not just in terms of quantity, but more importantly in terms of depth of research. In particular, the understanding of economic and business model issues surrounding OSS is limited (Feller, Finnegan, Kelly, & MacNamara, 2006b). In this section, we draw on the wider literature on business models and business networks to develop the theoretical grounding for our study. In particular, we examine how extant research on business models and networks can improve our understanding of the issues facing firms seeking to form the type of agile business ecosystems envisaged by Fitzgerald (2006).

In keeping with the increasing commercialization of OSS, researchers such as Krishnamurthy (2005), Weber (2004), Spiller and Wichmann (2002), Raymond (2001), and Hecker (2000) have documented a series of OSS business models. However, much of this work concentrates on the source of the revenue stream and neglects other aspects of the business models. This is not surprising, as the terms 'business model' and 'revenue model' are frequently, and incorrectly, used interchangeably. Looking outside the OSS literature, it is evident from the work of Timmers (1999), Mahadevan (2000), and Osterwalder and Pigneur (2002) that business models must examine value-adding activities in the context of a supply chain or business network. Osterwalder and Pigneur (2002) propose a comprehensive approach, and detail an ontology that focuses on four aspects

of the organization: product innovation, infrastructure management, customer relationship, and financials. Mahadevan (2000) defines a business model as a blend of three streams: value, revenue, and logistics. The value stream is concerned with the value proposition for buyers, sellers, and market makers. The revenue stream identifies how the organizations will earn revenue, and the logistics stream involves detailing how supply chain issues will affect the organizations involved. Timmers (1999) argued that architectures for business models can be identified through the deconstruction and reconstruction of a value chain. Value chain elements are identified, as are the possible ways that information can be integrated both within the value chain and between the respective value chains of interacting parties. Furthermore, Evans and Wurster (2000) argue that as more advanced information standards are introduced, levels of collaboration between organizations can be achieved that were previously only possible within a vertically integrated hierarchical intra-organizational structure. Thus, many economic entities have recognized the importance of the composition of the supply chain (or business network) to the overall performance of the firm (Christiaanse, 2005).

The benefits of cooperative business relationships have been advocated for decades (Kaufman, 1966; Van de Ven, 1976; Cash & Konsynski, 1985; Henderson, 1990; Finnegan, Galliers, & Powell, 2003). These relationships have been described as business webs (Tapscott, Ticoll, & Lowy, 2000), partnerships (Henderson, 1990), networks (Nelson, 1988; Joynt, 1991; Finnegan et al., 2003), strategic alliances (Joynt, 1991; Bronder & Pritzl, 1992; Lei & Slocum, 1992), virtual organizations (Davidow & Malone, 1992; Goldman, Nagel, & Preiss, 1995), joint ventures (Kanter, 1989; Oliver, 1990; Campell, Hollingsworth, & Lindberg, 1991), service consortia and stakeholder or value-chain partnerships (Kanter, 1989), promotional and obligation networks (Campell et al., 1991), agency federations, trade associations, social service joint programs, corporate-financial

interlocks, and agency-sponsor linkages (Oliver, 1990).

The reasons for such business cooperation include: resource procurement and allocation (Galaskiewicz, 1985; Clemons & Row, 1992; Alter & Hage, 1993), political advantages (Galaskiewicz, 1985), risk sharing and acquiring expertise (Alter & Hage, 1993), stability (Oliver, 1990), legitimacy (Galaskiewicz, 1985; Oliver, 1990), efficiency (Oliver, 1990; Clemons & Row, 1992), and innovating (Ticoll, Lowy, & Kalakota, 1998). Participants in business networks believe that collaboration will result in adaptive efficiency—the ability to change rapidly while providing customized services or products at a low cost (Alter & Hage, 1993). Thus, the ability to quickly assess new business opportunities, to identify suitable trading partners, and to effectively coordinate delivery of products and services across the business network is important (Sadeh, Hildum, & Kjenstad, 2003). Following this logic, agility is seen as an important characteristic of business networks.

Agility is a business-wide capability that includes organizational structures, information systems, logistics processes, and mindsets (Christopher, 2000). The term agility has created significant interest in the business world (Lo, 1998) and is recognized as a prerequisite for success in dynamic or turbulent environments (Christopher, 2000; Camarinha-Matos, Afsarmanesh, & Rabelo, 2003). Fingar (2000) believes that “the ability to change is now more important than the ability to create... Change becomes a first class design goal and requires business and technology architectures whose components can be added, modified, replaced and configured” (p. 66). Sharifi and Zhang (1999) argue that the concept of agility has two main attributes: responding to change promptly and appropriately, and capitalizing on the opportunities that are created by change. In a business network, agility is highly dependent not only on the skills of the individual firms, but also on the flexibility of the supporting infrastructure (Camarinha-Matos et al., 2003). Infrastructure flexibility has been identified as an important

characteristic of the agile organization (Christopher, 2000), and is dependent on both internal and external factors (Thomke & Reinertsen, 1998). External factors include changes in the needs of the end customer, while internal factors include changes in the development process. Thomke and Reinertsen (1998) argue that design flexibility can be brought about by: (1) following a development strategy that can endure a higher probability of design changes, (2) having the ability to produce late changes to the product design in order to better integrate it with the technology and the needs of the customer, and (3) preventing late changes to the product design by making design commitments at a very late stage in the development process.

To conclude, agile business networks designed to meet customer demands for customized products are reasonably well understood outside the OSS domain. Within the OSS field, recent work by Woods and Guliani (2005) and Fitzgerald (2006) has drawn attention to the importance of agile networks to OSS business models. However, small OSS firms have emerged from various OSS communities and thus cannot be considered to be the same as the type of firms that have been the subject of research on business networks to date. Therefore, there is a need for further research on the development of agile bazaar-friendly business ecosystems to deliver whole products in the OSS domain.

RESEARCH OBJECTIVE AND METHOD

The objective of this study is to explore the emerging phenomenon of a business network of OSS firms cooperating to deliver the 'whole product'. Two research questions were formulated to support this objective:

RQ1: *How does an OSS network affect the business models of participant organizations?*

RQ2: *What challenges are faced in ensuring that the network is agile?*

Case studies are regarded as the most commonly used qualitative research method in IS and are especially useful for studying organizational aspects of IS (Benbasat, Goldstein, & Mead, 1987). Cases are most appropriate when the objective involves studying contemporary events, without the need to control variables or subject behavior (Yin, 1994). The single case study method is considered to be a potentially rich and valuable source of data, while suited to exploring relationships between variables in their given context (Yin, 1994; Benbasat et al., 1987). We thus adopted a "soft positivist" epistemology as discussed by Kirsch (2004), and our method follows in the tradition of Eisenhardt (1989) and Madill, Jordan, and Shirley (2000); it is designed to reveal preexisting, relatively stable, and objectively extant phenomena and the relationships among them in a manner that is not limited to examining only pre-identified constructs.

The subject of the case study (Zea Partners) was chosen as it represented an interesting case in the area of open source business practice in that it is one of a small number of such networks aiming to deliver the 'whole product' in an OSS environment. The researchers first conducted a thorough archival search to determine the existence of public domain material on the network and participant companies. As a result of this preliminary analysis, the researchers prepared a case study protocol (cf. Yin, 1994). Based on this protocol, 16 interviews took place with key personnel from participant firms over a 17-month period from November 2004 to April 2006. In addition, the researchers had five separate interviews with the network founder (elite interviewing, cf. Marshall & Rossman, 1989) and also participated in four intensive workshops with network members during this time which facilitated member checking. The choice of interviewees was based on a number of factors. These were:

1. **Willingness to cooperate:** In order to obtain useful material, it was necessary for a potential interviewee to be interested in the study and willing to cooperate.

2. **History of network involvement:** Interviewees had to have been involved in ongoing network planning and/or project activity over a period of time. A consequence of this selection criterion was that the views of recent members were not studied.
3. **Seniority:** In order to get contextual material on business strategy and experience with network activities, it was necessary to speak with senior staff within each partner firm. A consequence of this selection criterion was that the views of junior staff were not studied.

Interviews, conducted using an interview guide (cf. Patton, 1980) were generally of one- to two-hour duration, with follow-up telephone interviews used to clarify and refine issues that emerged during transcription. Interviews were complemented by comprehensive reviews of documents and presentations at the workshops. The content analysis was conducted using Osterwalder and Pigneur's (2002) business model framework as well as Aitken et al.'s (2002) and Lee's (2004) agility frameworks. This is in line with Lee and Baskerville (2003), who, in addressing the issue of generalization, describe the process of generalizing from theory to empirical description (whereby the research seeks to apply findings confirmed in one setting to another one).

Case Environment

Zea Partners was founded in 2003 as the Zope Europe Association (ZEA) and changed its name to Zea Partners in 2006. Headquartered in Belgium, Zea Partners operates as an international network of businesses that build software and deliver services around the application server technology called Zope, which is widely used for developing content management systems, intranets, portals, and related applications. Zea Partners consists of 19 firms, with three managing partners and 16 associate partners located in The Netherlands, Italy, Norway, Belgium, Germany, the United Kingdom, Lithuania, the United States, Spain, France, and South Africa. The management team seeks project contracts

on behalf of network members and performs network management activities such as marketing and project management. The team also develops the network's business strategy in conjunction with the managing partners.

The partner companies are typically small (10 people or less). These companies have recognized that their size limits the contract (deal) sizes for which they could effectively compete, as well as their geographic range. One of the benefits of the network is, thus, that a number of companies can pool their resources to compete for larger contracts on a global scale. More importantly, in the context of competing on the basis of a whole product, the network allows partners to offer a full range of value chain activities, rather than concentrating exclusively on their own specialties (e.g., development, consultancy, training, etc.). The network is currently working on ensuring that all partners can conduct marketing under the one brand.

According to the network's founder, the goal is:

...to say that we have the whole product. We are going to group together all the people who need a whole product made but can't invest the resources to do it, and then take that whole product and make it offerable by anyone in the network. It has so many benefits on profitability it's just amazing. It's really the only way to impact profitability.

He sees this as being the value proposition of the network, and acknowledges that, through partnering, the network can compete for larger deal sizes without competing directly with the large international consulting companies. In comparing the Zea Partners network with such consulting companies, he notes the increased flexibility offered to customers. In particular, he argues that:

Instead of having a cathedral¹ model of Accenture, or something like that, we want to have multiple players in multiple countries. We can

move things around as new trends emerge, new specialties emerge, stuff like that."

He also highlights the importance of the fact that:

The people in the network are the people that created Silva, the creators of Plone, the creators of Workflow, the creators of Multilingual, the creators of each one of these things. And we want to explain to customers that it's in their interests to have a relationship that rewards these people. It's in their interests, first, because the guy who wrote it can get the job done at a pretty effective rate.

However, due to the early stage of development at which the network finds itself, coordination amongst partners is still on a person-to-person basis. There is an acknowledged need amongst members to evolve the organization of the network towards the use of quotas, geographical regions, and so forth. To date, coordination has meant observing trademark and domain rules, as well as some network terms and conditions to ensure that products/services delivered by partners meet the expectations of the customer. Finally, he acknowledges that it is critical to build trust amongst partners so that invoices are paid on time and other responsibilities are met.

Findings

The Osterwalder and Pigneur business model ontology was used as a lens to investigate how the presence of the Zea Partners network affects the business models of the member firms. The results are summarized in Table 1, classified as per the pillars of the Osterwalder and Pigneur (2002) framework, and are discussed below.

Zea Partners enhances the value that member firms can offer to a specific *target customer segment (value proposition)* by allowing smaller organizations to group together to deliver the whole product as part of a consortium. The fact that the network spans many geographic territories with multiple languages and specialized local knowledge means that a consortium

made up of small organizations can compete with the larger consultancy firms. This cooperation increases the range of projects in which members can become involved. Zea Partners covers 12 different countries which, from a geographical spread, makes it comparable with a large company. In terms of targeting customers, the Zea Partners brand is purposefully designed to be a mark that distinguishes participants in the network as being leaders in the market. Thus, organizations must already have a good reputation before they can join the network. The Zea Partners network also enhances the business model of participants by adding to the range of *capabilities* that underpin their value propositions. Many of the Zea Partners members are small start-ups that consist of two to four people with mostly specialized technical expertise. A major benefit of Zea Partners membership has, thus, been the ability to access Zea Partners' expertise in areas such as project management, customer relationship management, requirements management, tendering, and sales to complement technical expertise. According to the founder of one of the participant companies (Infrae), the development of *capabilities* within Zea Partners is vital to delivering a professional service. In relation to the production of documentation, he noted that "over time we gradually removed all dependencies on the community, because it was completely unpredictable."

With regard to *information strategy*, Zea Partners' stated aim is to "learn together, share experiences and refer leads to each other." However, as yet, resource problems have limited Zea Partners' ability to meet their ideal in terms of information strategy. Nevertheless, Zea Partners are proving very successful in enhancing the manner in which a participant firm reaches its customers (*feel and serve*). The market for Zope and Plone is characterized by customers approaching firms in the network with whom they want to do business. A key value added by Zea Partners is that the profile-building activities of the network results in 'leads' for member firms. In some countries where the demand for Plone services exceeds

Table 1. Effects of Zea Partners network on participants' business models

Business Model Pillar		Effect of Zea Partners on Members' Business Models
Product Innovation	Target Customer Segment	Enhances reputation and branding of participants by providing a single 'market leader' brand
	Value Proposition	Extends geographic coverage, supports the ability to offer specialized expertise, products and services in many languages, and leveraging local knowledge
	Capabilities	Enhances existing capabilities by providing a broader range of business capabilities, especially project management and customer relationship management
Customer Relationship	Information Strategy	Aims to provide lead referrals and to contribute to the sharing of experiences and knowledge
	Feel and Serve	Facilitates profile building through common branding
	Trust and Loyalty	Leverages access to expertise of software originators to build customer trust
Infrastructure Management	Resources	Lowers friction when building teams, through information sharing, common methodology, tracking results, reporting bugs, and so forth
	Activity Configuration	Enables members to act as a "value shop" configuration; network reduces the information asymmetry between client and consultant resulting in customers 'joining' the community
	Partner Network	Network means that members do not have to outsource to partners outside the network
Financials	Revenue Model	Increases deal size for members by creating "whole product" consortia
	Cost Structure	Enables cost-sharing amongst members
	Profit/Loss	Increases revenue and lowers expenses through sharing among members, leading to bigger profits

supply, Zea Partners can partner with member firms in that country by co-signing the deal but leaving the local participant with ownership of the customer. Furthermore, a key aspect of customer relationships amongst open source firms is that *trust and loyalty* can be enhanced by providing access to the originator of the software. Thus, Zea Partners aims to assemble project teams that contain relevant software originators from participant firms.

The OSS network model necessitates the inter-organizational management of business infrastructure. A key challenge has been integrating different participants in a seamless manner to deliver the 'whole product' to customers.

Zea Partners aims to lower friction in inter-organizational teams by establishing a common approach through the use of *resources*, for example: standing contracts, having customer references on file, having a common methodology or a common way of thinking about a problem, assigning work, tracking results, and reporting bugs. This is summed up by the Zea Partners founder as being the:

...big difference between a rabble and an army. You can take a thousand people that speak different languages, that never worked together and they can get defeated by 50 people that are well trained.

This approach is also evident in the Zea Partners approach to the *configuration of activities* and processes at the level of individual firms and at the network level. The Zea Partners network allows members to act as a 'value shop' (cf. Stabell & Fjeldstad, 1998) or service provider and carry out the phases of this configuration (problem finding and acquisition, problem solving, choice, execution, control, and evaluation) as if they were one integrated organization. One of the areas where Zea Partners differs from the "value shop" concept (as per Stabell & Fjeldstad, 1998) is that in the traditional "value shop" model, the information asymmetry between the client and the service provider (in this case a consultancy firm) is one of the main value drivers and results in high prices. This is not seen as desirable by Zea Partners. Instead the network endeavors to reduce this asymmetry so that the customer, instead of being a recipient of content management, becomes a participant in the OSS community:

There are certain people that need support contracts. There has not been a need for it in any of the projects we have been involved in. I'm a big believer in teaching the people to know enough about the solution to mostly fix it themselves. (Chief Architect, Plone Solutions)

Finally, in relation to infrastructure management, the *partner network* aspect of the business model is a service that Zea Partners completely operates on behalf of its members. Thus they do not need to outsource activities to non-member organizations.

Enhancing the *financial aspects* of the business models of participants is a key objective of the Zea Partners network. Zea Partners aims to increase the 'deal size' that members can tender for leading to increased profit margin. A key aspect of the Zea Partners' approach is, thus, the sharing of resources and common expenses. Thus participants can focus on key value-adding activities of their business models, while sharing the resources, costs, and risks of secondary value activities. An interesting revenue model arising in relation to OSS companies providing

consulting services is an effort to move away from a 'bill-by-the hour' model to fixed price. This move is occurring as the constant innovation with OSS results in the need for much less customization and thus shorter development times. However, fixed-price billing creates challenges for network-based project management as time overruns cut into the profit margins of the participant providing the service.

The discussion of Table 1 above refers to the effects of the Zea Partners network on the business models of the network participants, and not the business model(s) of Zea Partners itself. However, the various effects, taken as whole, result in the Zea Partners network operating as an entity in its own right, and engaging in agile competitive practices. Aitken et al. (2002) present a framework for understanding agility in the context of internal activities such as marketing, production, design, organization, management, and people. This framework is utilized in the present study as a tool for describing the agile characteristics of Zea Partners, as summarized in Table 2 and discussed below.

The members of Zea Partners are already independently able to utilize recognition of the Zeo and Plone brands as a marketing tool, but this is only relevant to client firms already aware of Zeo/Plone. The unified brand image of Zea Partners promotes agile marketing in several ways, such as simplifying brand management (one brand vs. many) and allowing the network to devote resources to unified brand building, reducing the burden on individual members (e.g., in order to increase brand awareness of Zea Partners and its member organizations, the founder is active in giving interviews, attending conferences, etc.). The long-term goal of Zea Partners is to build up sufficient resources so that the network can project a professional image on behalf of the member organizations that simply would not possess the resources to do this individually.

The most important characteristic of Zea Partners vis-à-vis production is its ability to leverage the large amount of diverse skills possessed by the member organizations. According to the chief architect of Plone Solutions:

Table 2. Agile characteristics of Zea Partners

Activity Area	Key Characteristics
Marketing	Network provides and maintains unified Zeo and Zea Partners brand
Production	Network provides harmonized and integrated collection of diverse production processes and capabilities to deliver the whole product
Design	Network provides harmonized and integrated collection of diverse design processes and capabilities to deliver the whole product
Organization	Network serves as competency rallying mechanism to deliver multi-lingual, whole product services across a wide geographic area
Management	Network distributes responsibility and revenue through simple, decentralized, and transparent network governance structures
People	Network provides customers with access to original software authors and/or experts with unique competencies

The thing that will make the network strong is that there is no single point of failure; you can swap out components or companies. If one company does not have the domain knowledge we normally have another company...It's very agile and very flexible.

Thus, the network allows delivery of the 'whole product', which would not be possible for the smaller members to do as a stand-alone provider. Likewise, many design issues associated with delivering the whole product are addressed through leveraging common experience with a common set of tools, working practices, communication norms, and cultures that serve to harmonize and integrate the practices of individual firms.

In terms of organizational activity, while the network facilitates matching member competency with customer need, Zea Partners does not currently use explicit coordination processes to schedule work. Previous attempts to do so have had negative results—for example, an incident in which a member firm was advised not to accept new work for a certain time period based on a client's intention, only to find that the client organization was unable to sign the contracts in the agreed timeframe, thus trapping the member firm into a period of non-productivity. However, by not having explicit scheduling mechanisms in place, Zea Partners believes

that it is more agile than traditional consulting firms, as the network is able to allocate resources more dynamically and effectively and thus to smooth out the peaks and valleys that are a characteristic of technology consulting and development work.

From a management perspective, the goal of Zea Partners is to ensure that network management and governance does not impede realizing the potential benefits associated with the fact that open source software is by nature highly decentralized. The founder of Zea Partners believes that this fact results in "a higher velocity of innovation," and that firms in the open source space are thus better equipped to adapt to the very specific needs of clients. For example, while a larger proprietary software development firm may decide not to support a particular language because the market is not big enough to sustain it, open source firms can leverage the work of individual developers and smaller groups who wish to support that language. Having a decentralized governance/management structure and a decentralized approach to consulting means that if a need is encountered for an unanticipated skill set, it is less of a problem to meet the need than it would be in the traditional consulting model. Finally, in relation to people, Zea Partners leverages the availability of access to the original author(s) and/or core maintainer(s) of the software

products to respond to customer demand in an agile fashion.

Zea Partners is made up of a number of autonomous organizations, each having different philosophies, operating in different countries, and meeting the needs of a diverse group of clients—meaning that it can draw from a wider variety of experiences. The chief architect of Plone noted that there are consequential social and management challenges:

You will, of course, get the complexity that comes from coordinating different companies with different working styles, and the whole chemistry thing where not all people have worked with the other people all the time.

The business agility challenges that are faced by the network were analyzed using the work of Lee (2004) as a lens. He expands on the concept of internal agility by also considering the adaptability of a supply chain as well as the alignment of players within that supply chain. Lee's work is used to frame the content analysis of agility-related challenges facing Zea Partners and is summarized in Table 3. The categorization of challenges according to this classification reveals that the challenges extend beyond agility to matters of adaptability and alignment. Consequently, while Zea Partners has been able to engage in agile competitive practices, the challenges that management has articulated indicate the need to move beyond short-term agile practices to consider structural and technological changes in OSS markets, and create performance incentives.

CONCLUSION

This article has responded to the need to expand our understanding of economic and business aspects of the OSS phenomenon (cf. Feller et al., 2006b) by exploring the business model and agility aspects of participation by open source companies in a business network designed to deliver the 'whole product' (cf. Fitzgerald, 2006). This participation is seen by those studied as a business imperative in order for small OSS firms to compete for large 'deal

sizes' with traditional integrated companies. The study indicates that participation in the network allows small firms to, in effect, outsource some elements of their business model to the network. This is particularly evident in the division of responsibility for customer-facing activities between participants and the central network. It is this division of responsibility that results in the network being able to engage in agile competitive practices, as network-level changes can be made rapidly in response to changes in the external environment. Thus, the challenges that the network faces in ensuring that the multitude of reciprocal interdependencies necessary for the delivery of a whole product do not adversely affect the agility of the network. Nevertheless, it is clear that agile practices are only the first step for the network in competing in the software and consulting sector. It is evident that further work is necessary to address adaptability and alignment issues.

The need for agility, adaptability, and alignment is a problem in all business networks. However, Zea Partners is not typical of other business networks, which rely on formal coordination mechanisms and legal agreements. Rather we observe that the relatively informal characteristics, found in the Zea Partners network, reflects the informal structures characteristic of the online communities of OSS developers from which the firms emerged. Reconciling these two approaches to the coordination issue is a critical issue for future research if OSS networks are to achieve adaptive efficiency (cf. Alter & Hage, 1993) and to deliver whole products in a 'bazaar-friendly' manner (Fitzgerald, 2006).

Overall, our study contributes to the understanding of the commercialization of open source software. Previous studies of commercial firms have been dominated by studies of single firms, whether OSS start-ups such as RedHat and JBoss (e.g., Krishnamurthy, 2005; Watson, Wynn, & Boudreau, 2005) or very large multinationals like Apple, IBM, and Sun (e.g., West, 2003). Our study examines the perspective of small/micro firms engaged in a cooperative business network in a manner that takes more complete consideration of the business model

Table 3. Key challenges for Zea Partners

AGILITY	
Objectives: To respond to short-term changes in demand or supply quickly and to handle external disruptions smoothly	Key Challenges: • Coordinate information flow amongst network participants to 'smooth out peaks and valleys' associated with traditional work • Foster collaborative relationships with partners based on the need for particular competencies • Develop network-level competencies (e.g., project management) to complement the core activities of participants
ADAPTABILITY	
Objectives: To adjust the network's design to meet structural shifts in markets; to modify supply network to strategies, products, and technologies	Key Challenges: • Leverage partner expertise in different geographical regions to understand the market for the total product • Plan for the introduction of new members into the network to meet requirements for particular competencies; also, ensure an adequate evaluation of potential members • Create an understanding of the needs of different types of customers (typically niche markets that traditional competitors do not serve) • Effectively manage the expertise of network partners to ensure that the competencies of the network evolve in response to changes in the product technologies that originate outside the network
ALIGNMENT	
Objectives: To create incentives for better performance	Key Challenges: • Exchange information and knowledge freely amongst network partners • Manage partner responsibilities in delivering the whole product in a manner that allows partners to focus on their core competencies • Effectively provide non-core competencies in a manner that participants can confidently delegate important business model components to the network • Equitably share risks, costs, and gains of initiatives • Enable customers to understand the business value of engaging with and contributing to the OSS community

concept than has been done to date in the OSS domain. Nevertheless, the methodology utilized for the study was exploratory, and thus the findings need further investigation. This study should be duplicated as part of the process of validating its findings in a context that is not just exploratory. In particular, further research is needed to replicate the study by assessing the results in a wider variety of networks.

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ENDNOTE

- ¹ Raymond (2001) first articulated the much cited contrast between the hierarchical cathedral model characterizing proprietary software and the distributed model of the open source bazaar.

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